

The many faces of Schnorr: a touch-up

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Abstract

In a previous paper [Shoup 2023], we presented a modular toolkit for analyzing threshold Schnorr signature schemes, built around “enhanced attack modes” in the non-distributed setting. In this note, we revisit the random oracle analysis of the combination of batch randomness extraction with re-randomized presignatures. We make the simulation argument fully explicit — in particular, giving a careful analysis of what is leaked through the presignatures, the programmed hash values, and the signature values — and fill in a gap in the treatment of batch re-randomization. We also give a simpler path to tight security bounds in the combined random oracle plus generic group model (ROM+GGM): rather than reducing to somewhat convoluted (if concrete) assumptions about the hash function within the GGM, we reduce in the ROM to an attack on the Schnorr interactive identification scheme, and then observe that the identification scheme is easily seen to be hard in the GGM.

1 Introduction

Threshold Schnorr signing protocols employ several techniques to achieve both efficiency and security. **Re-randomized presignatures** [GS21] mitigate presignature-based attacks by applying a random shift (from a random beacon) to each presignature at signing time. **Batch randomness extraction** [BHK⁺24, GS23] produces many “biased” presignatures by taking different linear combinations of “initial” presignatures, as specified by a “bias matrix” supplied by the adversary. **Batch re-randomization** [GS23] amortizes both the cost of invoking the random beacon and the scalar multiplication needed to compute shifted presignatures by sharing a single random shift across all signing queries in a batch.

The combination of these techniques is critical to the security analysis of the high-throughput threshold Schnorr schemes in [GS23] and [Sho26]. In [Sho23], a random oracle analysis was given for the combination of these techniques (Section 4.2 of that paper). However, that treatment had two shortcomings.

First, the claim that the hash values programmed by the simulator are uniformly and independently distributed was not fully justified. A careful argument must account for the joint distribution of the initial presignatures, the programmed hash values, and the signature values. Moreover, the adversary chooses the bias matrix *after* seeing the initial presignatures, introducing an adaptive dependence. We make all of this precise via an explicit matrix/vector equation (similar in spirit to the analysis in [Sho25]).

Second, the claim that the analysis “easily adapts” to batch re-randomization was compressed to a single sentence, overlooking a collision issue: when a single shift is shared across a batch, two signing queries with the same biased presignature and the same message would require the simulator to program the random oracle at the same point to two different values. We show that such a collision is either infeasible (because the simulator can extract a discrete logarithm) or entirely harmless (because the programmed values turn out to be identical).

Beyond filling these gaps, we give a simpler path to tight security bounds in the ROM+GGM. In [Sho23], the ROM (random oracle model) was used to reduce the signature attack to an attack on the Schnorr interactive identification scheme, which can then be reduced to the discrete logarithm problem via a forking-lemma argument, giving a loose bound. To get tight bounds, a direct but elaborate GGM (generic group model) analysis was carried out, reducing to somewhat convoluted (if concrete) preimage assumptions about the hash function. Here, we make the simple observation that if all one wants is a ROM+GGM security bound, there is a much more straightforward path: the ROM reduction to the identification scheme is already tight, and the identification scheme is easily shown to be hard in the GGM by a short, direct argument.

Organization. Section 2 sets up notation for Schnorr signatures, the identification scheme, and states a helpful lemma. Section 3 defines the combined attack game. Section 4 gives the ROM reduction. Section 5 proves the identification scheme secure in the GGM.

2 Preliminaries

2.1 Schnorr signatures

Let E be an elliptic curve over $\mathbb{Z}_p$ with a distinguished generator $\mathcal{G}$ of prime order q . The secret key is $d \xleftarrow{\$} \mathbb{Z}_q$ and the public key is $\mathcal{D} = d\mathcal{G} \in E$. The scheme uses a hash function $H : \{0, 1\}^* \rightarrow \mathbb{Z}_q$ and a prefix-free serialization $\langle \cdot \rangle$.

To sign a message m , one picks $r \xleftarrow{\$} \mathbb{Z}_q$, computes $\mathcal{R} \leftarrow r\mathcal{G}$, $h \leftarrow H(\langle \mathcal{D} \rangle \parallel \langle \mathcal{R} \rangle \parallel m)$, and $z \leftarrow r + h d$. The signature is $(\mathcal{R}, z)$. Verification checks $z\mathcal{G} = \mathcal{R} + h\mathcal{D}$.

2.2 Schnorr interactive identification

In the Schnorr interactive identification scheme, a prover with secret key d and public key $\mathcal{D} = d\mathcal{G}$ interacts with a verifier. In each session, the prover sends a commitment $\mathcal{R} = r\mathcal{G}$ for random r , receives a random challenge $c \xleftarrow{\$} \mathbb{Z}_q$ from the verifier, and responds with $z = r + c d$. The verifier accepts if $z\mathcal{G} = \mathcal{R} + c\mathcal{D}$.

We consider the **concurrent attack game**, in which an adversary may open many sessions concurrently (playing the role of prover), and wins if any verifier accepts. We denote the adversary’s advantage by $\text{IDadv}[\mathfrak{A}]$.

We note that in this version attack game, the adversary does not interact with an honest prover.

2.3 Adaptive uniformity

The following simple observation was made in [CS01], Lemma 9 (see also [CS03, Lemma 6.7]).

Lemma 2.1 (Adaptive uniformity). *Let $n \geq k \geq 1$ be integers and K a finite field. Let $\alpha \in K^n$ be uniformly random, and define $\beta = M\alpha + \gamma \in K^k$, where $M \in K^{k \times n}$ and $\gamma \in K^k$, with the i -th rows of M and γ determined by $\beta_1, \dots, \beta_{i-1}$. Then, conditioning on any values of $\beta_1, \dots, \beta_{k-1}$ such that the resulting matrix M has rank k , the value β_k is uniformly distributed over K .*

3 The combined attack game

We define the attack game that combines batch randomness extraction, re-randomized presignatures, and batch re-randomization. Let $P \leq Q$ be fixed parameters.

Attack Game 3.1 (CMA-BRE attack). *An adversary $\mathfrak{A}$ interacts with a challenger as follows.*

- The challenger computes $d \xleftarrow{\$} \mathbb{Z}_q$, $\mathcal{D} \leftarrow d\mathcal{G}$, and gives $\mathcal{D}$ to $\mathfrak{A}$.
- $\mathfrak{A}$ issues a series of queries, each of the following three types:

Presignature query: *In the k -th such query, the challenger generates a batch of **initial presignatures** $\mathcal{R}_{k,1}, \dots, \mathcal{R}_{k,Q}$ as random elements of E and gives them to $\mathfrak{A}$.*

Biasing query: *The adversary specifies a batch index k (for which initial presignatures have been generated but no previous biasing query has been made) and a bias $(U_k, \mathbf{u}'_k)$, where $U_k \in \mathbb{Z}_q^{P \times Q}$ is a full-rank matrix and $\mathbf{u}'_k \in \mathbb{Z}_q^P$. The initial presignatures are converted to **biased presignatures** $\bar{\mathcal{R}}_{k,1}, \dots, \bar{\mathcal{R}}_{k,P}$ via*

$$\begin{pmatrix} \bar{\mathcal{R}}_{k,1} \\ \vdots \\ \bar{\mathcal{R}}_{k,P} \end{pmatrix} = U_k \begin{pmatrix} \mathcal{R}_{k,1} \\ \vdots \\ \mathcal{R}_{k,Q} \end{pmatrix} + \mathbf{u}'_k \mathcal{G}. \quad (1)$$

Signing query (batch): *The adversary submits a batch of signing requests, each of the form $(k, i, m_{k,i})$, where k indexes a biased batch, $i \in [P]$, and each pair (k, i) is used at most once (across all signing queries). The challenger generates a single $\delta \xleftarrow{\$} \mathbb{Z}_q$ for the entire batch, and for each request computes the **effective presignature***

$$\mathcal{R}'_{k,i} := \bar{\mathcal{R}}_{k,i} + \delta \mathcal{G}$$

and the signature $(\mathcal{R}'_{k,i}, z_{k,i})$ on $m_{k,i}$ using $\mathcal{R}'_{k,i}$. The challenger returns δ and all signatures to $\mathfrak{A}$.

At the end, $\mathfrak{A}$ outputs $(m^, \mathcal{R}^*, z^*)$. We say $\mathfrak{A}$ **wins** if this is a valid signature on m^* and m^* was not submitted in any signing query.*

Remark 3.1. *When each signing batch contains a single request, we recover the (non-batched) re-randomized presignature game from [Sho23]. When $P = Q$ and $U_k = I_Q$, $\mathbf{u}'_k = \mathbf{0}$, we recover the setting without batch randomness extraction.*

4 Random oracle reduction

We show how to reduce the security of the combined attack game to the security of the Schnorr interactive identification scheme, modeling H as a random oracle.

4.1 The simulation strategy

The key is to simulate signing queries without knowledge of the secret key d . For each presignature batch k and each $j \in [Q]$, the simulator generates $\zeta_{k,j}, \eta_{k,j} \xleftarrow{\$} \mathbb{Z}_q$ and sets

$$\mathcal{R}_{k,j} \leftarrow \zeta_{k,j} \mathcal{G} - \eta_{k,j} \mathcal{D}. \quad (2)$$

These are uniformly distributed elements of E and have the correct distribution.

When the adversary specifies the bias $(U_k, \mathbf{u}'_k)$ with $U_k = (\sigma_{k,i}^{(j)})_{i \in [P], j \in [Q]}$ and $\mathbf{u}'_k = (\mu_{k,i})_{i \in [P]}$, we define

$$\bar{\zeta}_{k,i} := \mu_{k,i} + \sum_{j \in [Q]} \sigma_{k,i}^{(j)} \zeta_{k,j}, \quad (3)$$

$$\bar{\eta}_{k,i} := \sum_{j \in [Q]} \sigma_{k,i}^{(j)} \eta_{k,j}, \quad (4)$$

so that $\bar{\mathcal{R}}_{k,i} = \bar{\zeta}_{k,i} \mathcal{G} - \bar{\eta}_{k,i} \mathcal{D}$.

To process a signing batch with shared shift δ , the simulator chooses $\delta \xleftarrow{\$} \mathbb{Z}_q$ (once for the batch), and for each request $(k, i, m_{k,i})$ in the batch outputs the signature $(\mathcal{R}'_{k,i}, z_{k,i})$ along with δ , where

$$z_{k,i} := \bar{\zeta}_{k,i} + \delta, \quad (5)$$

and programs the random oracle so that

$$H(\langle \mathcal{D} \rangle \parallel \langle \mathcal{R}'_{k,i} \rangle \parallel m_{k,i}) := \bar{\eta}_{k,i}. \quad (6)$$

One verifies consistency: $z_{k,i} \mathcal{G} = (\bar{\zeta}_{k,i} + \delta) \mathcal{G} = \mathcal{R}'_{k,i} + \bar{\eta}_{k,i} \mathcal{D}$.

4.2 The matrix argument

We must argue that the joint distribution of the initial presignatures $\mathcal{R}_{k,j}$, the programmed hash values $\bar{\eta}_{k,i}$, and the signature values $z_{k,i}$ is correct, and that random oracle programming succeeds with overwhelming probability.

Fix an initial presignature batch k . Write $r_{k,j} = \zeta_{k,j} - d \eta_{k,j}$ for the discrete logarithm of the initial presignature $\mathcal{R}_{k,j}$. The information the adversary can derive about batch k is captured by the following linear system:

$$\underbrace{\begin{pmatrix} 1 & & & -d & & \\ & 1 & & & -d & \\ & & \ddots & & & \ddots \\ & & & 1 & & -d \\ & & & & \sigma_{k,1}^{(1)} & \sigma_{k,1}^{(2)} \cdots \sigma_{k,1}^{(Q)} \\ & & & & \vdots & \vdots \vdots \vdots \\ & & & & \sigma_{k,P}^{(1)} & \sigma_{k,P}^{(2)} \cdots \sigma_{k,P}^{(Q)} \end{pmatrix}}_{M_k} \begin{pmatrix} \zeta_{k,1} \\ \vdots \\ \zeta_{k,Q} \\ \eta_{k,1} \\ \vdots \\ \eta_{k,Q} \end{pmatrix} + \begin{pmatrix} 0 \\ \vdots \\ 0 \\ \mu_{k,1} \\ \vdots \\ \mu_{k,P} \end{pmatrix} = \begin{pmatrix} r_{k,1} \\ \vdots \\ r_{k,Q} \\ \bar{\eta}_{k,1} \\ \vdots \\ \bar{\eta}_{k,P} \end{pmatrix}. \quad (7)$$

Here, blank entries in the matrix are zero. The first column vector on the left has $2Q$ entries, the $\zeta_{k,j}$'s and $\eta_{k,j}$'s, which are uniformly and independently distributed over $\mathbb{Z}_q$.

We claim that $z_{k,i}$ leaks no additional information about the $\zeta_{k,j}$'s and $\eta_{k,j}$'s beyond what is already captured by (7). Indeed,

$$\bar{\zeta}_{k,i} = \mu_{k,i} + \sum_j \sigma_{k,i}^{(j)} \zeta_{k,j} = \mu_{k,i} + \sum_j \sigma_{k,i}^{(j)} (r_{k,j} + d \eta_{k,j}) = \mu_{k,i} + \sum_j \sigma_{k,i}^{(j)} r_{k,j} + d \bar{\eta}_{k,i},$$

so $z_{k,i} = \bar{\zeta}_{k,i} + \delta$ is determined by $(r_{k,1}, \dots, r_{k,Q})$, $\bar{\eta}_{k,i}$, δ , and the bias parameters.

Lemma 4.1. *The $(Q + P) \times 2Q$ matrix M_k in (7) has rank $Q + P$.*

Proof. Consider a linear combination of the rows that produces the zero vector, with coefficients $(a_1, \dots, a_Q, b_1, \dots, b_P)$. Looking at the first Q columns (the ζ -block), we get $(a_1, \dots, a_Q) = \mathbf{0}$ since the lower-left block is zero. Now looking at the last Q columns (the η -block), the contribution from the first Q rows vanishes, giving $(b_1, \dots, b_P) U_k = \mathbf{0}$. Since U_k has rank P , this forces $(b_1, \dots, b_P) = \mathbf{0}$. $\square$

Since M_k has full rank and the input vector $(\zeta_{k,1}, \dots, \zeta_{k,Q}, \eta_{k,1}, \dots, \eta_{k,Q})^T$ is uniform over $\mathbb{Z}_q^{2Q}$, the outputs on the right-hand side of (7) — the $r_{k,j}$'s and $\bar{\eta}_{k,i}$'s — are uniformly and independently distributed over $\mathbb{Z}_q$. Now, the first Q rows of M_k are fixed, while the last P rows are chosen by the adversary after seeing the initial presignatures $\mathcal{R}_{k,1}, \dots, \mathcal{R}_{k,Q}$ (i.e., after seeing $r_{k,1}, \dots, r_{k,Q}$). By Lemma 2.1, despite this adaptive choice, the $\bar{\eta}_{k,i}$ values remain uniform and independent of each other and of the initial presignatures, conditioned on the adversary's view.

Remark 4.2. *In fact, the adaptivity here is milder than the full generality of Lemma 2.1: the first Q rows of M_k are fixed, and after seeing the corresponding Q outputs, the adversary chooses the bias matrix U_k , which determines the remaining P rows. All P additional outputs are then determined simultaneously. Lemma 2.1 certainly applies to this setting.*

Since the input vectors are independent across batches k , the $\bar{\eta}_{k,i}$ values are also independent across different batches.

4.3 Success of random oracle programming

We must show that programming the random oracle at the points $(\langle \mathcal{D} \rangle \parallel \langle \mathcal{R}'_{k,i} \rangle \parallel m_{k,i})$ does not conflict with prior queries or create internal conflicts.

No conflict with prior queries. The effective presignature $\mathcal{R}'_{k,i}$ depends on δ , which is chosen uniformly *after* the adversary commits the signing batch. So $\mathcal{R}'_{k,i}$ is uniformly distributed from the adversary's viewpoint, and the probability that any single programming point was previously queried is at most N_h/q , where N_h is the number of random oracle queries made by the adversary. Summing over all N_{sig} signing requests across all batches, the total failure probability from this source is at most $N_{\text{sig}} N_h/q$.

No conflict within a signing batch. Since all requests in a batch share the same δ , two requests (k_a, i_a, m_a) and (k_b, i_b, m_b) in the same batch yield the same programming point if and only if $\bar{\mathcal{R}}_{k_a, i_a} = \bar{\mathcal{R}}_{k_b, i_b}$ and $m_a = m_b$ (the $\delta \mathcal{G}$ terms cancel). Suppose this occurs with $(k_a, i_a) \neq (k_b, i_b)$. The equality $\bar{\mathcal{R}}_{k_a, i_a} = \bar{\mathcal{R}}_{k_b, i_b}$ means $\bar{\zeta}_{k_a, i_a} \mathcal{G} - \bar{\eta}_{k_a, i_a} \mathcal{D} = \bar{\zeta}_{k_b, i_b} \mathcal{G} - \bar{\eta}_{k_b, i_b} \mathcal{D}$. There are two cases:

- If $\bar{\eta}_{k_a, i_a} \neq \bar{\eta}_{k_b, i_b}$, the simulator can extract the discrete logarithm as

$$d = \frac{\bar{\zeta}_{k_a, i_a} - \bar{\zeta}_{k_b, i_b}}{\bar{\eta}_{k_a, i_a} - \bar{\eta}_{k_b, i_b}}, \quad (8)$$

and can then trivially win the identification game.

- If $\bar{\eta}_{k_a, i_a} = \bar{\eta}_{k_b, i_b}$ (which happens with probability $1/q$), then also $\bar{\zeta}_{k_a, i_a} = \bar{\zeta}_{k_b, i_b}$ (from the group element equality). So the simulator programs the same point to the same value $\bar{\eta}_{k_a, i_a} = \bar{\eta}_{k_b, i_b}$ and produces the same signature value $z_{k_a, i_a} = z_{k_b, i_b}$. No conflict arises.

Thus, within-batch collisions either allow the simulator to extract d (and win), or are entirely harmless.

Remark 4.3. *In practice, it may be sensible for a protocol to de-duplicate messages within a signing batch, in which case within-batch collisions cannot occur (even if the biased presignatures coincide, distinct messages yield distinct programming points). However, the above analysis shows that de-duplication is not required for security.*

4.4 The reduction

The above simulation yields a reduction to the identification scheme. The reduction $\mathfrak{B}$ receives the public key $\mathcal{D}$ from the identification challenger and runs $\mathfrak{A}$, simulating all queries as described above. If a within-batch collision allows the extraction of d via (8), $\mathfrak{B}$ can trivially respond to any identification challenge. Otherwise, when $\mathfrak{A}$ produces a forgery $(m^*, \mathcal{R}^*, z^*)$ on a message m^* not submitted in any signing query, $\mathfrak{B}$ translates the forgery into a successful identification session by embedding the identification challenge into the appropriate random oracle response. This is the standard technique in the ROM analysis of Schnorr (see, e.g., Chapter 19 of [BS23]).

Theorem 4.4. *For any adversary $\mathfrak{A}$ against the CMA-BRE attack game (Attack Game 3.1) in the random oracle model, there exists an efficient adversary $\mathfrak{B}$ against the concurrent Schnorr identification scheme, with running time about the same as that of $\mathfrak{A}$, such that*

$$\text{CMA-BREadv}[\mathfrak{A}] \leq \text{IDadv}[\mathfrak{B}] + O(N_{\text{sig}} N_h / q),$$

where N_{sig} is the total number of signing requests across all batches and N_h is the number of random oracle queries.

Proof sketch. The reduction proceeds as described in Sections 4.1–4.3. The simulation is perfect except with probability $O(N_{\text{sig}} N_h / q)$ from random oracle programming failures (conflicts with prior queries). As shown in Section 4.3, within-batch collisions either allow

the extraction of d (in which case $\mathfrak{B}$ trivially wins the identification game) or are harmless. The forgery is translated into a successful identification session by the standard Fiat-Shamir-to-ID reduction. $\square$

Remark 4.5. *The above analysis extends straightforwardly to handle **additive key derivation**, where the adversary specifies a tweak $e_{k,i} \in \mathbb{Z}_q$ in each signing query and the effective public key becomes $\mathcal{D}'_{k,i} = \mathcal{D} + e_{k,i} \mathcal{G}$. The only change is that the effective public key is included in the hash, and the z -value is adjusted by the tweak. The matrix argument is unaffected.*

Remark 4.6. *The fixed parameter P in [Attack Game 3.1](#) could be replaced by a variable parameter P_k , allowing a different number of biased presignatures to be produced from each batch of initial presignatures. This models batch randomness extraction protocols that may yield a variable number of biased presignatures. The analysis is unchanged.*

5 Security of Schnorr identification in the GGM

We now show that the Schnorr interactive identification scheme is secure in the generic group model, with a tight $O(N^2/q)$ bound. Combined with [Theorem 4.4](#), this gives tight security for the full combined attack game in the ROM+GGM. Note that [\[Sho97\]](#) already analyzed the Schnorr interactive identification scheme in the GGM, but in a slightly different context (the attack game as well as the GGM formal model being slightly different); for completeness, we sketch a proof here.

We work in the EC-GGM of [\[GS21, Sho23\]](#). A random encoding function $\pi : \mathbb{Z}_q \rightarrow E$ is chosen; the adversary interacts with a group oracle that computes $\pi(c_1 \pi^{-1}(\mathcal{R}_1) + c_2 \pi^{-1}(\mathcal{R}_2))$ for points $\mathcal{R}_1, \mathcal{R}_2 \in E$ and scalars $c_1, c_2 \in \mathbb{Z}_q$ chosen arbitrarily by the adversary.

Theorem 5.1. *Any adversary making at most N group oracle queries against the concurrent Schnorr identification scheme in the EC-GGM wins with probability at most $O(N^2/q)$.*

Proof. Following the proof technique of [\[GS21\]](#), we replace the real game by a **symbolic simulation**, in which the domain of π is replaced by the set of all degree-one polynomials $a + b\mathcal{D} \in \mathbb{Z}_q[\mathcal{D}]$, where $\mathcal{D}$ is a formal indeterminate representing the discrete logarithm d , and $a, b \in \mathbb{Z}_q$ are known to the simulator. The adversary sees $\pi(1)$ (encoding $\mathcal{G}$) and $\pi(\mathcal{D})$ (encoding $\mathcal{D}$), and may compute arbitrary $\mathbb{Z}_q$ -linear combinations via group oracle queries.

The transition from the real game to the symbolic simulation incurs an error of $O(N^2/q)$: two distinct polynomials $a + b\mathcal{D}$ and $a' + b'\mathcal{D}$ evaluate to the same value at the secret d with probability $1/q$ (for $(a, b) \neq (a', b')$). A union bound over all $\binom{N}{2}$ pairs gives the claimed error.

In the symbolic simulation, consider a single identification session. The adversary submits a commitment, whose preimage under π is some $a + b\mathcal{D}$, where a and b are already fixed at the time the commitment is submitted. It then receives a random challenge $c \xleftarrow{\$} \mathbb{Z}_q$. To make the verifier accept, the adversary must produce $z \in \mathbb{Z}_q$ satisfying

$$z = (a + b\mathcal{D}) + c\mathcal{D},$$

as formal polynomials. Matching the constant term gives $z = a$; matching the D-coefficient gives $b + c = 0$. Since b is fixed before c is drawn uniformly, this holds with probability exactly $1/q$. A union bound over all sessions completes the proof. $\square$

Combining [Theorems 4.4](#) and [5.1](#):

Theorem 5.2. *In the ROM+GGM, any adversary making at most N total queries (to the signing, group, and random oracles) against the CMA-BRE attack game wins with probability $O(N^2/q)$.*

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